

Analyzing Self-Reported Data Collection Methods for Clinical Intake

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Fig. 1: A self reporting form showing a Likert scale.

Abstract—The rapid evolution of artificial intelligence tools has reshaped data handling, offering new paradigms for data collection. Our work provides a comparative analysis of three data collection methods for clinical intake: traditional online forms, conversational AI chatbots, and AI-enhanced online forms. We conducted a within-subjects user study utilizing persona cards to simulate realistic clinical intake scenarios, preceded by a preliminary heuristic evaluation. Our findings reveal distinct trade-offs: traditional forms provide high user control and efficiency; chatbots offer natural interactions but suffer from verbosity and reduced navigability; and AI-enhanced forms present a highly effective middle ground, combining the speed of traditional forms with the intelligent guidance of AI. We also discuss limitations in automated error-rate calculations and provide actionable design guidelines for implementing AI in sensitive self-reporting contexts.

Index Terms—Data Collection, Self Reporting Forms, AI Chatbot, AI enhancements, Clinical Intake

1 INTRODUCTION

Clinical intake relies heavily on self-reported patient data, including symptoms, medications, and medical history. The accuracy of this data is critical, as it directly impacts diagnosis and treatment decisions [1]. However, existing data collection methods often introduce cognitive challenges for users. Traditional online forms, while highly structured, can be rigid and filled with medical terminology that patients struggle to interpret, leading to incomplete or inaccurate responses [3].

Recent advances in large language models have enabled conversational interfaces that allow users to describe their conditions in natural language. These chatbot systems have shown promise in increasing engagement and reducing the cognitive load associated with interpreting forms [6]. However, they also introduce usability concerns, including limited visibility of system status, reduced user control, and inefficiency for simple data entry tasks.

To address the limitations of both extremes, a third paradigm has emerged: AI-enhanced forms. These systems retain the familiar, highly navigable structure of traditional forms while embedding targeted, automated features such as inline clarification and real-time translation.

The rationale for comparing these three specific paradigms is to understand the spectrum of automation integration in data collection. We contrast a baseline with no artificial intelligence (Traditional), a system

where a language model completely dictates the interaction flow (Chatbot), and a hybrid system where automation acts as a supplementary tool controlled by the user (AI-Enhanced). Our research question is: *How do varying levels of AI integration impact the efficiency, usability, and user trust of clinical data collection interfaces?*

2 RELATED WORK

Prior work in Human-Computer Interaction regarding data collection can be categorized into three primary themes: the optimization of traditional forms, the rise of conversational agents, and the emerging space of hybrid interfaces.

Usability in Structured Data Collection: Historically, research has extensively explored the design of usable web forms. Applying strict usability guidelines, such as logical grouping of fields, immediate validation feedback, and clear typography, can significantly reduce user errors and improve completion time [2, 7]. These findings highlight that users highly value structured interfaces that allow them to visually parse the entire scope of a task before beginning.

The Shift to Conversational Interfaces: Conversational interfaces have emerged as an alternative to static forms, particularly in healthcare. Studies demonstrate that chatbot-based surveys can elicit more detailed, empathetic responses compared to traditional forms [4]. Users often perceive these interactions as more natural and less akin to a chore [9]. However, this naturalness comes with steep trade-offs. Conversational systems fundamentally restrict user control, forcing a linear progression that makes it difficult to review, edit, or cross-reference previous inputs [10].

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Trust and Hybrid Systems: More recent work has begun to explore hybrid approaches that combine structured interfaces with adaptive assistance. These systems aim to preserve the reliability and auditability of forms while introducing adaptive support. Research indicates that users maintain higher levels of trust in automated systems when they retain a sense of agency and control over the final output [5, 8]. Our work builds directly upon this literature by evaluating these three paradigms empirically within a unified, clinical intake scenario.

3 SYSTEM DESIGN

To effectively evaluate our research question, we developed a comprehensive custom web application built using the React.js framework for the frontend client and a Node.js runtime environment for the backend server architecture. This application was designed to simulate a realistic patient portal experience. We created three distinct interface prototypes for a clinical intake questionnaire, as illustrated in Figure 2. To ensure experimental validity, each interface was strictly constrained to collect the exact same nine fields of information from the user. These fields included standard demographic data such as Name and Date of Birth, as well as complex medical inquiries including Pain Level, Known Allergies, Current Medications, and Primary Reason for Visit.

3.1 Traditional Online Form

The traditional form serves as our baseline control condition, representing the current industry standard for digital clinical intake. It presents all nine input fields simultaneously in a standard, vertically scrolling web layout. Users interact with the system via manual text entry and standard HTML form controls like dropdown menus and number inputs. The navigational structure is entirely non-linear. This means users can freely jump between different input fields, skip questions they are unsure about to return to them later, and review all entered data prior to final submission. This design heavily prioritizes the Nielsen heuristics of user control and freedom, as well as visibility of system status, by allowing the user to view the entire scope of the data collection task at a single glance.

3.2 LLM-Powered Conversational Chatbot

To represent a fully automated, dialogue-driven interaction paradigm, we designed a sequential chatbot powered by the Anthropic Claude Haiku 4.5 language model. Instead of viewing a static page of empty boxes, users respond to the system dynamically in natural language. The underlying language model was strictly prompt-engineered to ask exactly one question at a time and strictly validate inputs before moving forward. For example, the system ensures that the reported pain level is a distinct integer from zero to ten, and it is programmed to reject nonsensical inputs like future dates for birthdays. While this interface provides a highly conversational and empathetic user experience, it intentionally obscures the overall system status. It forces a linear data-entry path, requiring the user to depend entirely on the system’s pacing rather than their own.

3.3 AI-Enhanced Online Form

The hybrid AI-enhanced form was developed to combine the static, predictable layout of the traditional form with on-demand intelligent assistance. This prototype was also powered by the Claude Haiku 4.5 model API. In this interface, users maintain complete navigational control over the document, but they can trigger language model tools for specific, individual fields. The system features a clarification function, which is designed to translate complex medical terminology into plain, accessible language for patients with lower health literacy. It also includes a translation function to assist non-native speakers in understanding exactly what a specific field requires. This design rationale relies heavily on the concept of selective automation. By providing intelligent assistance only when explicitly requested by the user, the system aims to preserve patient trust and data auditability while mitigating the cognitive load associated with medical data entry.

4 METHODOLOGY

4.1 Preliminary Heuristic Evaluation

Prior to conducting our empirical user study, we performed an expert review of the three software prototypes using Nielsen’s usability heuristics. We selected this specific evaluation framework because it provides a well-established set of principles for assessing interface usability across different interaction models. We specifically isolated five heuristics that are highly relevant to the unique, high-stakes challenges of clinical data collection.

Match between the system and the real world (H1) was included because clinical intake forms often require users to interpret dense medical terminology. Interfaces that align with a patient’s natural language and mental model reduce cognitive friction. User control and freedom (H2) is essential in self-reporting tasks where users may need to revise or correct their health inputs to prevent downstream medical errors. Visibility of system status (H3) was selected due to its importance in maintaining user confidence during data entry, ensuring patients understand their progress through what can be a lengthy intake questionnaire. Error prevention and recovery (H4) is incredibly relevant in clinical scenarios where incorrect inputs, such as wrong medication dosages, can lead to severe diagnostic issues. Finally, flexibility and efficiency of use (H5) was included to evaluate how easily users of varying technological literacy levels can complete the task without feeling burdened. This preliminary analysis provided a theoretical baseline for understanding the structural trade-offs of each interface paradigm before exposing them to live participants.

4.2 User Study Protocol

To evaluate the three interfaces empirically, we designed and conducted a within-subjects experimental user study. By having every participant test every interface, we were able to directly compare individual preferences and performance metrics across the paradigms.

Participants and Protocol: Participants interacted with all three system designs via our locally hosted web application. To prevent the collection of actual protected health information and to standardize the task difficulty across all sessions, participants were provided with fictional persona cards. These cards contained comprehensive patient profiles varying widely in condition complexity, patient age, and medical history. Participants were instructed to adopt the persona on the card and complete the intake process using the assigned interface as if they were that specific patient. The order in which the interfaces were presented to the participants was strictly counterbalanced. This method was implemented to mitigate learning effects and fatigue, ensuring that no single interface benefited simply from being the last one tested.

Measures: The application software was engineered to automatically record quantitative telemetry in the background during the testing sessions. This included precise task completion time measured in seconds and input error rates calculated by comparing the final submitted form to the original persona card data. Following the completion of each individual interface, participants filled out a post-task survey. We measured self-reported usability and trust using a standard 5-point Likert scale, where 1 indicated strong disagreement and 5 indicated strong agreement. Finally, participants provided open-ended qualitative feedback regarding their experience, allowing us to capture nuanced opinions that telemetry alone could not reveal.

5 RESULTS AND DISCUSSION

5.1 Preliminary Evaluation Insights

The initial heuristic evaluation revealed that no single paradigm dominates across all usability dimensions (Table 1). The chatbot inherently matched natural conversation (H1), but severely limited user control (H2) and system visibility (H3). The traditional form excelled in visibility and control but provided limited error prevention (H4). The AI-enhanced form emerged theoretically as the most balanced approach, preserving the navigational freedom of traditional forms while introducing intelligent error prevention and assistance.

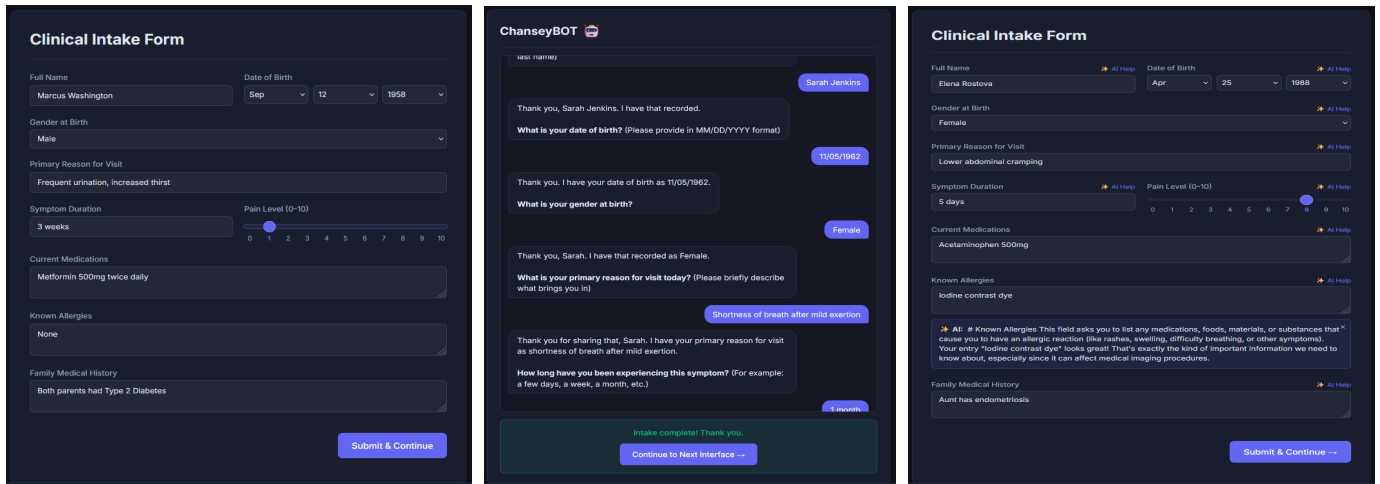


Fig. 2: The three interface paradigms for clinical intake data collection. **Left:** Traditional online form with static fields. **Center:** LLM-powered conversational chatbot with adaptive questioning. **Right:** AI-enhanced form combining structured layout with per-field AI assistance.

Table 1: Preliminary heuristic ratings for each interface paradigm evaluated across five key Nielsen heuristics.

Heuristic	Traditional	AI-Enhanced	Chatbot
H1: Real World Match	Medium	High	Very High
H2: User Control	High	Medium	Low
H3: Status Visibility	High	Medium	Low
H4: Error Prevention	Low	Very High	Medium
H5: Flexibility	High	High	Low

5.2 Quantitative Analysis

Completion Time: Telemetry data revealed clear differences in task efficiency across the three experimental conditions (Figure 3). The Traditional form was generally the fastest method for data entry. Because all input fields were simultaneously visible on a single page, users could leverage their preexisting familiarity with standard web interfaces to rapidly scan, type, or copy information. Conversely, the Chatbot interface was consistently the slowest, significantly increasing the total completion time for every participant. This delay was primarily driven by the conversational system’s strict sequential pacing, which forced users to wait for the language model to generate each subsequent prompt and frequently required them to construct full-sentence responses rather than inputting simple, discrete data points. The AI-Enhanced form served as a highly effective middle ground. While it was marginally slower on average than the purely Traditional form—owing to the fact that users occasionally paused to interact with the optional AI clarification features—it remained vastly faster than the rigid conversational constraints of the Chatbot. This indicates that intelligent assistance can be integrated without a devastating cost to basic efficiency.

Usability and Trust Scores: Analysis of the 5-point Likert scale responses (Figure 4) indicated a remarkably strong user preference for the AI-Enhanced form. The Traditional form scored adequately on baseline usability metrics due to its widespread familiarity, but lacked the supportive scaffolding that users desired when confronted with complex medical inputs. In contrast, the AI-Enhanced form consistently achieved the highest combined scores for both Usability and Trust. Participants felt supported by the system’s intelligence without losing their sense of individual agency or control over their personal data. The Chatbot, however, saw highly polarized usability scores, reflecting a distinct divide in user preferences regarding conversational interfaces. While a subset of participants found the chatbot highly engaging and less intimidating, a significant portion penalized it heavily for its lack of navigational freedom, its verbosity, and the inability to quickly review

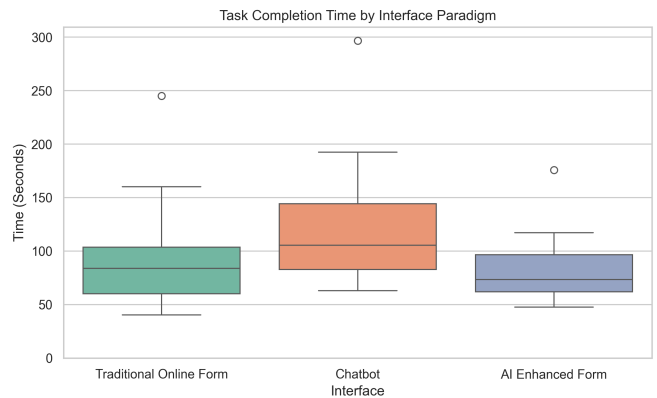


Fig. 3: Task completion time (in seconds) across the three interface paradigms. The Chatbot required significantly more time to complete the identical task.

or edit prior inputs.

Limitations of Error Rate Metric: It is necessary to note a methodological limitation regarding the quantitative error rate. The system calculated errors using a strict string-matching algorithm between the user’s input and the persona card. This approach was overly aggressive, flagging minor, expected human variations (e.g., capitalization, minor misspellings, or formatting differences like “10 mg” vs “10mg”) as critical errors. Furthermore, the web application lacked an anti-clipboard mechanism. Users frequently bypassed manual typing by copying and pasting directly from the persona cards, artificially deflating the true error rate. Consequently, the error rate data was deemed unreliable for comparative statistical analysis.

5.3 Qualitative Analysis

Thematic analysis of the open-ended user feedback provided crucial context for the quantitative telemetry, revealing four distinct themes that govern user interaction with automated data collection systems:

1. Conversational Comfort vs. Verbose Inefficiency: User sentiment regarding the Chatbot was heavily polarized, largely dependent on the user’s technical proficiency and mindset going into the task. Some users appreciated the conversational tone, noting that it reduced the clinical sterility of the process: *“The chatbot made it feel more comfortable and conversational. Less like a task.”* Conversely, users who approached the intake as a simple administrative chore found it frustratingly slow. As one participant stated: *“Bookkeeping systems*

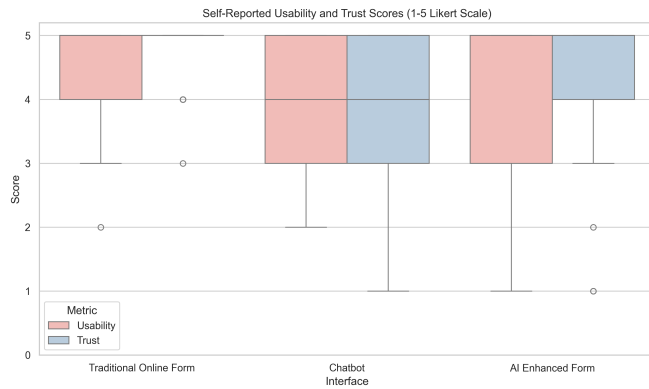


Fig. 4: Self-reported Usability and Trust scores across the three evaluated interface paradigms.

should be designed for efficiency and ease of use... I found the verbose nature of the interface to be redundant and wasteful." This highlights that while empathy is valued, it should not supersede basic usability in task-oriented interfaces.

2. Guidance and Expectation Setting: Users consistently appreciated when the interface clearly guided them on expected inputs, effectively lowering the anxiety often associated with medical terminology. Even in the highly constrained Chatbot condition, users noted that "Prompting lets you know what to expect," highlighting the intrinsic value of large language models in clarifying vague medical questions and helping users construct accurate responses.

3. The Efficiency of AI-Enhanced Forms: The hybrid approach was widely praised for successfully balancing the speed of a traditional interface with the intelligence of an automated system. Users enjoyed the empowerment of having tools available without being forced to use them. One user summarized the general consensus: "Personally, I ranked the AI-Enhanced Online form the highest out of three because I thought that it was the fastest compared to the Traditional Online Form and the AI Chatbot."

4. The Uncanny Valley of Over-Automation: Users were acutely aware of when automation was applied unnecessarily, which actively detracted from their trust in the system. In the AI-Enhanced form, providing language model assistance for universally understood fields caused cognitive friction. As one user pointedly noted, "Was weird to have an option for AI help for the full name." This highlights a crucial design guideline: intelligent enhancements should be strictly reserved for complex, ambiguous, or high-cognitive-load tasks to avoid alienating the user.

6 CONCLUSION

Our comparative evaluation demonstrates that while automation has immense potential to revolutionize clinical data collection, the architectural method of its integration dictates its ultimate success. Fully conversational chatbots, while highly natural and empathetic, fundamentally violate key usability heuristics regarding user control and system visibility. Forcing users into a rigid, sequential dialogue results in significantly slower completion times and high frustration levels during simple data entry tasks. Conversely, the Traditional form remains highly efficient for power users but severely lacks the support necessary for patients who may struggle with complex medical terminology or health literacy.

The AI-Enhanced form emerged definitively as the superior paradigm. By preserving the familiar, fully auditable structure of a standard web form and embedding on-demand assistance via Claude Haiku 4.5, we successfully balanced optimal efficiency with intelligent support. This approach allowed users to bypass automation when they felt confident, while providing robust safety nets for difficult questions. Future iterations of clinical intake software should aggressively adopt this hybrid approach. System designers must ensure that arti-

cial intelligence acts as an empowering, opt-in tool that augments the user's natural data entry process, rather than a restrictive gatekeeper that dictates it.

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